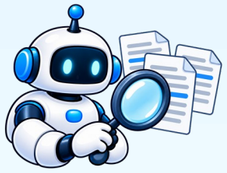
HyWe Framework

Search-based mode ($\mathcal{S}$)



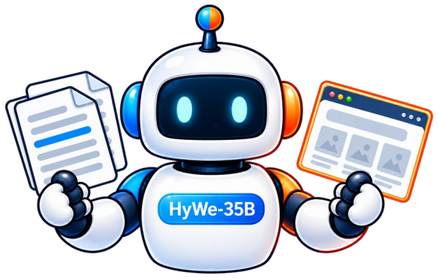
Role prompt

You are a general-purpose agent on the **SEARCH-BASED** path.
Your tools this turn: search, scholar, visit, code_sandbox, **SWITCH**.
- `SWITCH` hands control to the operation-based (browser) path. Use it only when the needed information requires real page interaction (login, form, dynamic content ...).

Available tools

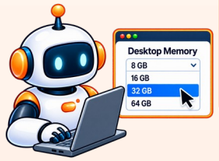
search | scholar | visit | code_sandbox

$\mathcal{S}$ action



$\mathcal{O}$ action

Operation-based mode ($\mathcal{O}$)



Role prompt

You are a general-purpose agent on the **OPERATION-BASED** path (real browser).
You act on the live page: goto, new_tab, click, ...
`SWITCH` hands control back to the search-based path.
Once a page is open you get its screenshot each turn - ground every action in what is actually visible there.

Available tools

goto | click | type | scroll

observation

SWITCH action

3 **shared context**
whole trajectory → model input

observation

Trajectory h_t 1 **unified action–observation interface**

S segment → *becomes shared history after SWITCH*

search(query) → *text* → visit(url) → *text*

2 **SWITCH**
swap role prompt & tools · new segment

O segment *(current)*

goto(url) → *screenshot* → click(x, y) → *screenshot*

HyWeCollector



Real websites (InSTA-150k)

- Product sites
- Knowledge sites
- Local services
- ...

Capability Probing

1. Task prototypes

36 search-base	48 operation-base
historical context reputation and reviews deep evidence synthesis ...	product configurator map route mortgage calculator ...

2. Switching patterns

- A_1 $\mathcal{O} \to \mathcal{S}$: switch for efficiency
- A_2 $\mathcal{S} \to \mathcal{O}$: switch when search fails
- A_3 $\mathcal{S} \to \mathcal{O}$: switch to next $\mathcal{O}$ subgoal
- A_4 $\mathcal{O} \to \mathcal{S}$: switch to next $\mathcal{S}$ subgoal

3. Multi-agent synthesis

Task synthesizer

1 **Instantiate**

blueprint $b = (\varphi, w, e, c, d)$
φ A3 : S → O
w egghead.io
e a free beginner React hooks course
c S_ENTITY_BY_CONSTRAINTS ∘ O_COURSE_FILTER
d easy

2 **Verify** ✗ retry

validators ✓ all pass
route S → O matches φ
phases alternate, 1 each
sites O phase binds a site
deps consumed value produced
N(b) task text hides the plan

✓ pass → $x = N(b), C(b)$

Task

Find the well-known free beginner React hooks course on egghead.io, open it, confirm it plays, and report its details.

Solution

$C(b) = \{(g_k, \delta_k, m_k)\}$
g_1 $\mathcal{S}$ identify the course
δ_1 title + URL resolved
↓ s1.result
g_2 $\mathcal{O}$ open, confirm playback
δ_2 playback verified

Executor

Use search / browser tools to execute tasks

executor brief
sees g_1 · verified results · feedback
hidden $C(b)$ · later goals · policies

Current Trajectory ↓ ↑ *Feedback*

Controller

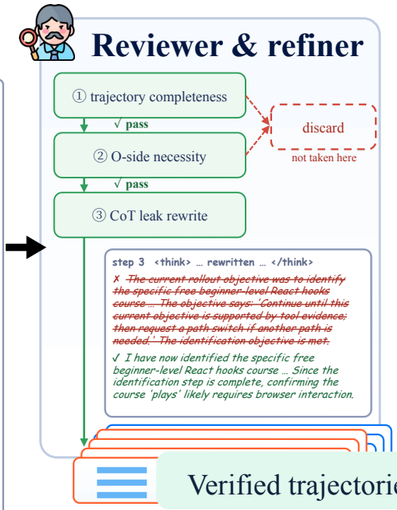
Check progress and gate SWITCH/ ANSWER

phase verdicts
switch @3 δ_1 met, ev 1-2 ✓
answer @8 δ_2 met, ev 4-7 ✓

Raw Hybrid Trajectory



Reviewer & refiner



Verified trajectories